

FOSSIL PLANTS

Plants are among the first organisms to appear in the fossil record. Algae are found in rocks from the Precambrian. Vascular plants (plants with tissues for conducting water and nutrients) evolved in the Silurian Period, and by Carboniferous times, Earth was coloured green by vast, coal-forming swamp forests. Flowering plants developed later, in the Mesozoic Era.

SEED FERN LEAF

Alethopteris serlii

A seed fern from Carboniferous and Permian strata, *Alethopteris* had compound pinnate fronds, consisting of thick, strongly veined leaflets.



CLIMBING HORSETAIL

Sphenophyllum emarginatum
Found fossilized in rocks of Carboniferous to Permian age, this horsetail had wedge-shaped leaves and long, soft stems adapted for climbing.

SIGILLARIA STEM

Sigillaria aevolaris

Found in Carboniferous and Permian rocks, *Sigillaria* was a giant relative of clubmosses that grew to over 30 m (100 ft). It had a narrow stem, and its leaves grew in clumps.



pinnate
(feather-
like) leaf

EARLY LAND PLANT

Cooksonia hemisphaerica

Found in Silurian and Devonian rocks, *Cooksonia* was one of the earliest vascular plants. It had a stiff stem and leafless branches.



CYCLOPTERIS LEAFLETS

Cyclopteris orbicularis

Oval leaflets from a seed fern called *Neuropteris* are given the scientific name *Cyclopteris*. Its fossils belong to Carboniferous strata.

SEED FERN SEEDS

Trigonocarpus adamsi

The name *Trigonocarpus* is given to fossil seeds from seed ferns found in Carboniferous strata. Each seed has three ribs.



CALAMOPHYTON STEMS

Calamophyton primaevum

A primitive, leafless plant, probably related to ferns, *Calamophyton* is found in Devonian and Early Carboniferous rocks.



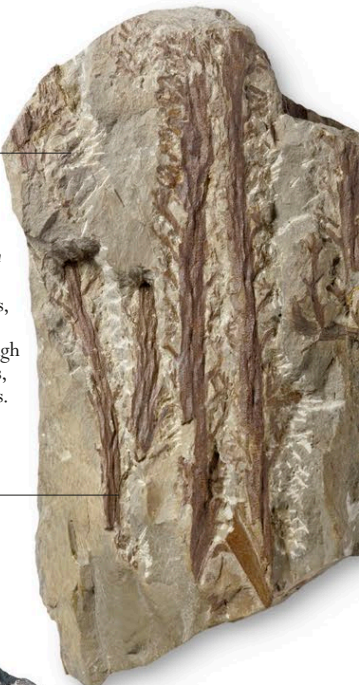
CLADOXYLON STEMS

Cladoxylon scoparium

Fossilized in Devonian and Carboniferous rocks, *Cladoxylon* was a low-growing plant, with a tough central stem and leafless, light-absorbing branches.

branch

tough stem



HORSETAIL FOLIAGE

Asterophyllites equisetiformis

Found in Carboniferous and Permian strata, *Asterophyllites* had needle-shaped leaves and a structure similar to that of modern horsetails.



LEPIDODENDRON ROOT

Stigmara ficoides

Occurring in Carboniferous to Permian rocks, *Stigmara* is the name for the fossil roots of the clubmoss relative, *Lepidodendron*.



SALVINIA RHIZOME

Salvinia formosa

A floating water fern from the tropics, *Salvinia* is found fossilized in rocks from Cretaceous to recent strata.



CRETACEOUS FERN

Weichselia reticulata

Found fossilized in Cretaceous strata, *Weichselia* was similar to modern bracken and had fronds that were divided twice.

SEED FERN LEAFLETS

Dicrodium sp.

A seed fern from the Triassic Period, *Dicrodium* had pinnate leaves and its fronds were about 7.5 cm (3 in) long.

